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5 November 1979
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MILITARY SPECIFICATION

TEST SET, CHEMICAL AGENT AUTOMATIC

ALARM: ABCA-M74

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes requirements for one type of field maintenance test set.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

FSC 6665

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter. :

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-252 - Wired Equipment, Classification of Visual and Mechanical Defects for.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

D5-15-5360 - Test Set, Chemical Agent Automatic Alarm: ABCA-M74.
D5-15-6553 - Packaging for Test Set, ABCA-M74.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D5-15-5360 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings and in this specification.

3.1.2 Components. All components of the test set shall conform to the specifications and drawings listed on Drawing D5-15-5360 and subsidiary drawings.

3.2 Manufacture and assembly. The components shall be manufactured and assembled in accordance with Drawing D5-15-5360 and the subsidiary drawings.

3.3 Continuity. The interconnecting wiring of the test set shall have a resistance of less than 1.0 ohm between the points listed in table I when tested as specified in 4.4.4.1.

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TABLE I. Continuity between points

From	To
CHASSIS TEST cable, pin A	TP1
CHASSIS TEST cable, pin B	TP2
CHASSIS TEST cable, pin C	TP3
CHASSIS TEST cable, pin D	TP17
CHASSIS TEST cable, pin D	CHASSIS TEST cable, pin F
CHASSIS TEST cable, pin D	CHASSIS TEST cable, pin I
CHASSIS TEST cable, pin D	CHASSIS TEST cable, pin W
CHASSIS TEST cable, pin E	TP4
CHASSIS TEST cable, pin G	TP5
CHASSIS TEST cable, pin H	TP6
CHASSIS TEST cable, pin J	TP7
CHASSIS TEST cable, pin K	TP8
CHASSIS TEST cable, pin L	TP9
CHASSIS TEST cable, pin N	TP10
CHASSIS TEST cable, pin O	TP11
CHASSIS TEST cable, pin P	TP12
CHASSIS TEST cable, pin Q	TP13
CHASSIS TEST cable, pin R	TP14
CHASSIS TEST cable, pin T	TP15
CHASSIS TEST cable, pin U	TP16
DETECTOR POWER cable, pin A	J2-A
DETECTOR POWER cable, pin B	PUMP POWER cable, pin C
DETECTOR POWER cable, pin B	TP19
DETECTOR POWER cable, pin C	J1-H
DETECTOR POWER cable, pin D	J1-Q
DETECTOR POWER cable, pin E	J2-E
DETECTOR POWER cable, pin E	J1-T
DETECTOR POWER cable, pin E	DETECTOR POWER cable, pin A
J4-M	TP21
J1-T	J4-D
J1-P	J1-W
J1-P	TP19
J1-P	J4-B (with S4 in SHUNT pos)
J1-P	TP22 (with S4 in SHUNT pos)
J1-P	TP18 AMP
J1-I	+E1
J2-A	-E2
J2-B	+E1
J2-E	TP20
J4-B	

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3.4 Resistance. The test set shall have the resistance specified in table II when tested as specified in 4.4.4.2.

TABLE II. Resistance between points

From	To	Resistance (ohms)
CHASSIS TEST cable, pin S	TP15	442,000 ± 5,000
CHASSIS TEST cable, pin T	CHASSIS TEST cable, pin V	887,000 ± 9,000
CHASSIS TEST cable, pin T	CHASSIS TEST cable, pin S	442,000 ± 5,000
PUMP POWER cable, pin A	J2-E	500 ± 100 with pot R-11 fully counter-clockwise
PUMP POWER cable, pin B	TP19	Open circuit, 1.0 max. when switch S5 is operated.
J4-M	TP22	30 maximum
J4-C	J4-B	30 maximum
J4-H	J4-B	680 ± 40
J1-A	J1-B	20,000 ± 2,000
J1-B	J1-C	100 max. with pot R10 fully counter-clockwise
J1-K	J1-L	Open circuit, with switch S3 in OPERATE position, 1.0 max. with switch S3 in RESET position.
J1-N	J1-O	Open circuit, with switch S3 in RESET position, 1.0 max. with switch S3 in OPERATE position.
PUMP POWER cable, pin A	DETECTOR POWER cable, P2-E	1.0 max. with VOLTAGE ADJUST pot fully clockwise.

3.5 Electrical functioning.

3.5.1 Meter circuits. The meter circuits shall function as required below when tested as specified in 4.4.4.3.

(a) With potentiometer R11 fully clockwise (as viewed from front of panel), switch S1 in the SUPPLY position, and a direct current voltage applied at J2-A and J2-B such that meter M1 indicates 24.0, the voltage

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measured between E1 and TP20 shall be 24.0 ± 1.0. When S1 is moved to the MOTOR VOLTAGE position, and the applied voltage adjusted such that meter M1 indicates 24.0, the voltage measured between E1 and TP20 shall be 24.0 ± 1.0.

(b) With switch S4 in the SHUNT position and 17.0 ± 0.1 volt direct current (vdc) applied between J1-U and J1-W, and switch S1 in the 17V TEST position, the indication of meter M1 shall be within the green zone.

(c) With switch S4 in the SHUNT position and a 90.5 ± 1.0 microampere direct current applied at J1-S and J1-W, and switch S1 in the METER CIRCUIT position, meter M1 shall indicate 40 ± 1 division.

(d) With a 90.5 ± 1.0 microampere direct current applied at J1-V and J1-I, and switch S1 in the AMP TEST position, meter M1 shall indicate 40 ± 1 division.

(e) With battery BT1 removed, switch S2 in the TEST position, switch S1 in the TRIGGER TEST position, and a direct current voltage applied at the battery terminals of the test set such that the pointer of meter M1 is at center of the red zone, a voltage of 15.0 ± 1.0 millivolt (mv) shall be present at J1-E and J1-G and the voltage applied at the battery terminals shall be less than 5.0 volts. When switch S2 is placed in the ZERO position, the 15.0 ± 1.0 mv present at J1-E and J1-G shall decrease to zero within 1.0 second.

3.5.2 ALARM ACTUATE switch circuits. The voltage present at J4-R and J4-P when the ALARM ACTUATE switch S6 is held in either position, shall be 5.0 ± 1.0 volt when 5.0 ± 1.0 volt is applied at the battery terminals of the test set when tested as specified in 4.4.4.4.

3.5.3 HORN CIRCUIT and REMOTE CIRCUIT lamp operation. With 24.0 ± 2.0 vdc applied to J2-A and J2-B (positive terminal of power source to J2-A), and switch S4 in the SHUNT position, the HORN CIRCUIT and REMOTE CIRCUIT lamps shall operate as follows when tested as specified in 4.4.4.5.

(a) With J1-R connected to J1-W the HORN CIRCUIT lamp only shall be energized.

(b) With J1-H connected to J1-W the REMOTE CIRCUIT lamp only shall be energized.

(c) With J1-Q connected to J1-T through a 2500 ohm 5 percent resistor, the REMOTE CIRCUIT lamp only shall be energized.

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(d) With J1-R and J1-H connected to J1-W, the HORN CIRCUIT lamp only shall be energized.

(e) With J1-R connected to J1-W and with J1-Q connected to J1-T through a 2500 ohm 5 percent resistor, the HORN CIRCUIT lamp only shall be energized.

(f) With J1-R and J1-H connected to J1-W and with J1-Q connected to J1-T through a 2500 ohm 5 percent resistor, both the HORN CIRCUIT and the REMOTE CIRCUIT lamps shall be energized.

3.5.4 Circuit card voltage regulator. With 24.0 ± 2.0 vdc applied to J2-A and J2-B, J4-E connected to J4-D, and J4-H connected to J4-B through a 25 ohm, 2 watt, 20 percent resistor, and a voltage of 5.5 ± 1.0 volt shall be present between J4-H and J4-B when tested as specified in 4.4.4.6.

3.5.5 INLET TEMP test circuit. With any direct current voltage between 18.0 (+1.0 or -0.0) vdc and 36.0 (+0.0 or -1.0) vdc applied to J2-A and J2-B, switch S1 in the INLET TEMP position, and the thermistor in the instrumented paddle assembly (Drawing D5-15-5534) at a temperature of 92.5° ± 0.5°F (36.6 ± .28°C), the indication of meter M1 shall be within the red zone when tested as specified in 4.4.4.7.

3.6 Pneumatic system functioning.

3.6.1 Pressure system. The hand-operated pressurizing device of the test set shall increase the pressure in the pressure system to at least 25.0 inches (63.5 cm) of water above ambient pressure. With the pressure system at a pressure of 24.0 ± 1.0 inch (60.96 ± 2.54 cm) of water above ambient pressure and closed-off from the pressurizing device, the rate of pressure change in the pressure system shall not exceed 0.25 inch (0.64 cm) of water per minute. With the pressure system at pressures of 12.0 ± 1.0 and 6.0 ± 1.0 inch (30.48 ± 2.54 and 15.24 ± 2.54 cm) of water above ambient pressure and closed-off from the pressurizing device, the PRESSURE-VACUUM GAGE shall indicate the true system pressure within 0.50 inch (1.27 cm) of water. Test shall be conducted as specified in 4.4.4.8.

3.6.2 Vacuum system. The pressure drop through the vacuum system shall be less than 5.0 inches (12.7 cm) of water at an airflow of 1.1 ± 0.2 liter per minute with the FLOW ADJUST control fully counterclockwise. With the test port of the flowmeter plugged, and the vacuum system at a pressure of 24.0 ± 1.0 inch (60.96 ± 2.54 cm) of water below ambient pressure and closed off from the low pressure source, the rate of pressure change within the vacuum system shall not exceed 0.25 inch (0.64 cm) of water per minute when tested as specified in 4.4.4.9.

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3.6.3 Flowmeter accuracy. When the test set flowmeter is tested as specified in 4.4.4.10 with the test set flowmeter connected in series with a sapphire ball float, variable-area, standard flowmeter (e.g., Fischer and Porter 1/8-16-G-5, Tri-Flat R), having a demonstrated accuracy of ± 1.0 percent, and with a flow rate through the flowmeters of 1.4 liters per minute as indicated by the test set flowmeter, the flow rate indicated by the standard flowmeter shall be 1.4 ± 0.1 liter per minute at ambient conditions of 75° ± 15°F (23.89 ± 8.33°C) temperature, less than 90 percent relative humidity, and 757.5 ± 12.5 millimeters (29.82 ± 0.49 inches) of mercury pressure.

3.7 Operation at 32°F (0°C). The test set shall meet the requirements of 3.5, 3.6.1, and 3.6.2 when the test set is at a temperature of 32° ± 5°F (0° ± 2.78°C) when tested as specified in 4.3.3.1.

3.8 Operation at 105°F (40.56°C). The test set shall meet the requirements of 3.5, 3.6.1, and 3.6.2 when the test set is at a temperature of 105° ± 5°F (40.56° ± 2.78°C) when tested as specified in 4.3.3.2.

3.9 Operation after vibration. The test set shall meet the requirements of 3.5 and 3.6 after exposure to the vibration environment described in table III. Vibration shall be sinusoidal, and applied along each of three mutually perpendicular axes of the test set. Test shall be conducted as specified in 4.3.3.3.

TABLE III. Test set vibration environment

Frequency range (Hertz)	Double amplitude (inches, ± 10%)	Sweep time (minutes, ± 10%)
5 to 15	0.06	2.5
15 to 25	0.04	2.5
25 to 55	0.02	3.0

3.10 Preproduction. Prior to the start of regular production, a preproduction sample of the test sets shall be produced in accordance with this specification for examination and test (see 4.3).

3.11 Workmanship. The workmanship of the test set shall be in accordance with MIL-STD-252.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of three test sets shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedures for examination and nondestructive tests. The preproduction sample of test sets shall be examined in accordance with 4.4.3.3 and tested in accordance with 4.3.3 and 4.4.4 for conformance to the requirements of this specification.

4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 Operation at 32°F (0°C). Place the test set, with the case lid open, into a temperature test chamber already at the temperature required by 3.7. Allow at least two hours for the test set to reach chamber temperature. Perform the tests described in 4.4.4.3 through 4.4.4.9 with the test set at the temperature required by 3.7. When the testing at this temperature is completed, return the test set to ambient temperature and again perform the tests described in 4.4.4.3 through 4.4.4.9. Do not permit moisture to condense on the test set while it is returning to ambient temperature.

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4.3.3.2 Operation at 105°F (40.56°C). Place the test set, with the case lid open, into a temperature test chamber already at the temperature required by 3.8. Allow at least two hours for the test set to reach chamber temperature. Perform the testing described in 4.4.4.3 through 4.4.4.9 with the test set at the temperature required by 3.8. When the testing at this temperature is completed, return the test set to ambient temperature and again perform the tests described in 4.4.4.3 through 4.4.4.9.

4.3.3.3 Operation after vibration. Mount the test set on the table of a vibration exciter in such a manner that the main panel of the test set will be perpendicular to the axis of vibration. Vibrate the test set as required by 3.9. Remount the test set, and vibrate it, along each of the other two mutually perpendicular axes of the test set. When the vibration required by 3.9 has been completed, perform the tests described in 4.4.4.3 through 4.4.4.10 and visually inspect for any physical damage and loose components or hardware.

4.3.4 Acceptance/rejection criteria. The test set shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the test sets offered for acceptance at one time, which have been produced by one manufacturer, at one plant, under essentially the same manufacturing conditions.

4.4.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedures.

4.4.3.1 For examination. Sample test sets and the preparation for delivery shall be examined in accordance with the classification of defects (4.4.3.3) and MIL-STD-105.

4.4.3.2 For test. Each test set in the lot shall be tested in accordance with 4.4.4. Any test set found defective shall be rejected.

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4.4.3.3 Classification of defects.

(a) Test set, chemical agent automatic alarm: ABCA-M74 (Drawing D5-15-5360).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Workmanship (applicable defects listed as Major in MIL-STD-252, see 3.11)
Minor:	AQL 4.0 percent defective	
201		Workmanship (applicable defects listed as Minor in MIL-STD-252, see 3.11)
	AQL 6.5 percent defective	
202		Workmanship (applicable defects listed as Control in MIL-STD-252, see 3.11)

(b) Packaging (section 5).

Categories		Defects
Critical:		None defined
Major:	AQL 2.5 percent defective	
101		Packaging or packing component missing, incorrect, or incorrectly assembled
102		Packaging or packing component damaged
103		Markings missing, incorrect, or illegible

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Continuity. Connect an ohmmeter having an accuracy of ± 2° of an arc on R X 1 scale, 1.5° of arc on the R X 100 and R X 10K scale successively to each pair of test points listed in table I. The resistance measured at each pair of test points shall be as required by 3.3.

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4.4.4.2 Resistance. Connect an ohmmeter having an accuracy of ± 2° of an arc on R X 1 scale, 1.5° of arc on the R X 100 and R X 10K scale successively to each pair of test points listed in table II. Establish any condition of measurement required by table II. The resistance measured at each pair of test points shall be as required by 3.4.

4.4.4.3 Meter circuits.

(a) Adjust the mechanical zero adjust of meter M1 such that the meter pointer is centered in the blue band. Place switch S1 in the SUPPLY position. Rotate knob of potentiometer R11 fully clockwise (as viewed from front of panel). Connect a suitable voltmeter (e.g., 50 VDC, 1% full scale) between E1 and TP20 with the meter plus terminal connected to E1. Connect a variable voltage (0-28) direct current source to J2-A and J2-B, with the plus terminal of the power source connected to J2-A. Increase the voltage applied at J2-A and J2-B until meter M1 indicates 24.0. At that time the voltage measured between E1 and TP20 shall be as required by 3.5.1(a). Place switch S1 in the MOTOR VOLTAGE position. Adjust the voltage applied at J2-A and J2-B such that meter M1 again indicates 24.0. At that time the voltage measured between E1 and TP20 shall be as required by 3.5.1(a). Disconnect the power source and the voltmeter from the test set.

(b) Place switch S4 in the SHUNT position and S1 in the 17V TEST position. Connect a source of 17.0 ± 0.1 vdc to J1-U and J1-W, with the plus terminal of the power source connected to J1-U. The indication of meter M1 shall be as required by 3.5.1(b). Disconnect the power source from the test set.

(c) Place switch S4 in the SHUNT position and S1 in the METER CIRCUIT position. Connect a 90.5 ± 1.0 microampere direct current source, monitored using a 100 microamp meter having an accuracy of 0.5 percent, to J1-S and J1-W, with the plus terminal of the power source connected to J1-S. The indication of meter M1 shall be as required by 3.5.1(c). Disconnect the power source from the test set.

(d) Place switch S1 in the AMP TEST position. Connect a 90.5 ± 1.0 microampere direct current source, monitored using a 100 microamp meter having an accuracy of 0.5 percent, to J1-V and J1-I, with the plus terminal of the power source connected to J1-V. The indication of meter M1 shall be as required by 3.5.1(d). Disconnect the power source from the test set.

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(e) Place switch S1 in the TRIGGER TEST position, and switch S2 in the TEST position. Connect a variable voltage (0.0 - 5.0) source of direct current to the battery terminals of the test set, with the plus terminal of the power source connected to the plus battery terminal. Connect a suitable millivoltmeter (e.g., 25 mv dc, 1% full scale) between J1-E and J1-G, with the plus terminal of the meter connected to J1-G. Slowly increase the voltage applied to the battery terminals until the pointer of meter M1 is at the center of the red zone. With the pointer of meter M1 at the center of the red zone, the applied voltage and indication of millivoltmeter shall be as required by 3.5.1(e). Place switch S2 in the ZERO position and observe that the voltage indicated on the millivoltmeter decreases to zero within the time specified. Disconnect the power source and the meter from the test set.

4.4.4.4 ALARM ACTUATE switch circuits. Connect a source of 5.0 ± 1.0 vdc to the battery terminals of the test set, with the plus terminal of the power source connected to the plus terminal. Connect a suitable voltmeter (e.g., 10 vdc, 1% full scale) between J4-R and J4-P. Measure the voltage present when the ALARM ACTUATE switch is moved to first one ON position, then to the other ON position. The voltage measured at J4-R and J4-P shall be as required by 3.5.2.

NOTE: The polarity of the voltage will change as the switch is operated.

4.4.4.5 HORN CIRCUIT and REMOTE CIRCUIT lamp operation. Place switch S4 in the SHUNT position. Connect a 24.0 ± 2.0 vdc power source to J2-A and J2-B, with the positive terminal of the power source connected to J2-A. Perform the following tests and observe the response of the REMOTE CIRCUIT lamp and HORN CIRCUIT lamp during each test. The response of the lamps shall be as specified in 3.5.3.

(a) Momentarily connect J1-R to J1-W.
(b) Momentarily connect J1-H to J1-W.
(c) Momentarily connect J1-Q to J1-T through a 2500 ohm 5 percent resistor.
(d) Momentarily connect both J1-R and J1-H to J1-W.
(e) Momentarily connect J1-R to J1-W and connect J1-Q to J1-T through a 2500 ohm 5 percent resistor.
(f) Momentarily connect both J1-R and J1-H to J1-W, and connect J1-Q to J1-T through a 2500 ohm 5 percent resistor.

Disconnect the power source from the test set.

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4.4.4.6 Circuit card voltage regulator. Connect J4-E to J4-D. Connect J4-H to J4-B through a 25 ohm, 2 watt, 20 percent resistor. Connect a suitable voltmeter (e.g., 10 vdc, 1% full scale) across the 25 ohm resistor. Connect a source of 24.0 ± 2.0 vdc to J2-A and J2-B, with the plus terminal of the power source connected to J2-A. The voltage indicated by the voltmeter connected across the resistor shall be as required by 3.5.4.

4.4.4.7 INLET TEMP test circuit. Place switch S1 in the INLET TEMP position. Connect a direct current source, continuously variable between 18 and 36 volts and capable of at least 0.5 ampere output, to J2-A and J2-B with the plus terminal of the power source connected to J2-A. Support the instrumented paddle assembly in a vessel of water at 92.5° ± 0.5° (33.61° ± 0.28°C) such that the visible end of the thermistor is at least 1/4 inch (0.64 cm) but less than 1/2 inch (1.27 cm) below the surface of the water. Turn on the power source and adjust the output to 24.0 ± 2.0 volts. If necessary, adjust R24 to cause the pointer of meter M1 to be centered in the red zone. Adjust the output of the power source to 18.0 (+1.0 or -0.0) volts, and note the indication of meter M1. While observing the indication of meter M1, increase the output of the power source to 36.0 (+0.0 or -1.0) volts. The indication of meter M1 shall stay within the red zone, as required by 3.5.5. Disconnect the power source from the test set. Carefully dry the instrumented paddle assembly before storing it in the test set.

4.4.4.8 Pressure system. Adjust the zero adjust of the PRESSURE-VACUM GAGE such that the gage pointer is centered in the zero graduation. Connect the hand-operated pressurizing device to the PRESSURE PORT. Rotate the knob of the PRESSURE LOCK to its fully counterclockwise position. Connect a suitable pressure measuring device (e.g., a water manometer with a range of at least 25 inches [63.5 cm]) to the end of the PRESSURE TEST line. Examine the VACUUM TEST line; remove any kinks or plugs that may be present. Using the hand-operated pressurizing device, increase the pressure in the pressure system until the PRESSURE-VACUUM GAGE indicates 25.0 inches (63.5 cm) of water. Rotate the knob of the PRESSURE LOCK to its fully clockwise position. Observe the rate of pressure change within the pressure system for at least 2.0 minutes; the rate of pressure change shall be as required by 3.6.1. Disconnect the pressurizing device. Rotate the knob of the PRESSURE LOCK counterclockwise sufficiently to permit the system pressure to decrease first to the 12.0 ± 1.0 inch (30.48 ± 2.54 cm) of water pressure and then to the 6.0 ± 1.0 inch (15.24 ± 2.54) of water pressure required by 3.6.1. At each of these pressures compare the indications of the PRESSURE-VACUUM GAGE and the pressure measuring devices at the end of the PRESSURE TEST line; these indications shall be as required by 3.6.1. Disconnect the pressure measuring device. Rotate the knob of the PRESSURE LOCK to its fully counterclockwise position.

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4.4.4.9 Vacuum system. Rotate the knob of the FLOW ADJUST control to its fully counterclockwise position. Connect the end of the VACUUM TEST line to the inlet port of an air pump having an airflow rate of 1.1 ± 0.2 liter per minute. This connection should include a means of isolating the air pump from the test set (e.g., a short length of surgical tubing and a pinch-clamp). Examine the PRESSURE TEST LINE, remove any kinks or plugs that may be present. Start the air pump and observe the PRESSURE-VACUUM GAGE. The indication of pressure drop on the PRESSURE-VACUUM GAGE shall be as required by 3.6.2.

CAUTION: Do not drive the PRESSURE-VACUUM GAGE off scale during the following test.

Insert a stopper into the detector flowmeter test port. When the pressure in the vacuum system has been reduced to 24.0 ± 1.0 inch (60.96 ± 2.54 cm) of water below ambient, close-off the air pump from the vacuum system. Stop the air pump. Observe the rate of pressure change within the vacuum system for at least 2.0 minutes; the rate of pressure change shall be as required by 3.6.2. Disconnect the air pump from the test set and remove the stopper from the flowmeter test port.

4.4.4.10 Flowmeter accuracy. Rotate the knob of the FLOW ADJUST control to its fully counterclockwise position. Connect the end of the VACUUM TEST line to the inlet port of an air pump having an airflow rate greater than 1.6 liters per minute. Connect the outlet of the standard flowmeter described in 3.6.3 to the detector flowmeter test port of the test set. Start the air pump and observe the test set flowmeter. Rotate the knob of the FLOW ADJUST control clockwise, if necessary, to produce the test flow rate required by 3.6.3. Note the flow rate indicated by the standard flowmeter; the indicated flow rate shall be as required by 3.6.3. Disconnect the air pump and the standard flowmeter from the test set, and return the FLOW ADJUST control to its fully counterclockwise position.

5. PACKAGING

5.1 Preservation and unit packing, level A. The test set shall be unit packed in accordance with Drawing D5-15-6553.

5.2 Packing. Level A or commercial (see 6.2).

5.2.1 Level A. The test set, unit packed as specified in 5.1, shall be packed in containers conforming to the following:

(1) Box, wood, class 2, style 4, grade B, type 1 load, weight of contents 0-50 pounds, of PPP-B-621.

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(2) Box, cleated plywood, overseas type, style optional, grade B, type 1 load, weight of contents 0-100 pounds, of PPP-B-601. The inside dimensions of the boxes shall be 23-1/8 by 20 by 12-3/4 inches (58.74 by 50.8 by 32.39 cm) (+1/8 inch [0.32 cm] tolerance on each of the dimensions) length, width, and depth, respectively. The boxes shall be closed and strapped in accordance with appendix of their respective specifications. Strapping and seals shall conform to QQ-S-781 [Strapping - type 1 (regular duty), finish B, grade 2, and Seals - type D, style I or IV, finish B, grade 2].

5.2.2 Commercial packing. A uniform quantity of test sets, unit packed as specified in 5.1, shall be prepared for shipment as specified in MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order, unit packs (5.1) shall be marked in accordance with MIL-STD-129 (interior unit pack markings). Level A shipping containers (5.2.1) shall be marked in accordance with Drawing D5-15-6564. Commercial shipping containers (5.2.2) shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. This test set is intended for use in the field maintenance of the ABCA-M43 detector unit and the ABCA-M42 alarm unit of the ABCA-M8 series chemical agent alarms.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(c) Level A or commercial packing required.

6.3 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 251. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization officer, if required.

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Preparing activity:

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